750 / 40 5N Red Gold

Colour: red Density: 15,0 g/cm^3 Melting Range: 880…900°C

Chemical Composition

Element	Au	Ag	Cu						
Amount [‰]	750	40	210						

Mechanical Properties

	casted, not deformed	**annealed, recrystallisated**	hardened, 300°C 60 min
Hardness [HV]	(160)	**160**	300
Tensile Strength R_m [N/mm^2]		**560**	950
Yield Strength $R_{p\,0,2}$ [N/mm^2]		**300**	850
Fracture Elongation [%]		**≥30**	<5

Annealing Properties

Furnace	Temperature	Annealing Time	Holding Time	Quenching Medium	Atmosphere
Continous Furnace, Batch Furnace	650°C	15 min			H_2/N_2-Mixture, max. 8 % H_2

Investment-casting :
Flasktemperature :600-640°C Casting Temperature : 1010-1040°C Quenching in water after 5min.(!!)
Cleaning acids : 20% Hydrofluoric acid or De-vesting from the supplier of investment-powder

